Day 046 | Unit 04 | Teacher plan

Numeric and Algebraic Expressions

Write and Evaluate Powers

Standards: 6.EE.A.1

Learning target: I can write repeated multiplication using a whole-number exponent.

enVision Topic 3: Numeric and Algebraic Expressions | Spectrum companion Chapter 5

0–5 minutes / Launch

Read the target. Ask for an estimate or a sketch before calculating. Connect the new idea to the previous lesson.

5–17 / Model

Use Worked Example 1, then Example 2. Ask students to name the next step and explain a choice. The base is the repeated factor and the exponent tells how many copies to multiply. For a positive exponent, 4^3 = 4 × 4 × 4, not 4 × 3. Parentheses show when a whole product is the base.

17–42 / Practice

Assign questions 1–10 in order. Use 1–3 for foundations, 4–7 for application, and 8–10 for state-format response and reasoning. The example button provides a separate question with a worked method.

42–50 / Discuss and check

Compare representations. Review a selected answer and a plausible error. Use question 10 as an exit discussion; ask students to check units, scale, or substitution.

Support and extension

Offer number lines, fraction strips, grid paper, or a table template as appropriate. Read multi-part directions one part at a time. For extension, ask for another representation or a counterexample.

Companion and state alignment

Use the matching skill in the edition you own. No unverified publisher page numbers are assigned. AASA references below identify related public sample items by standard and format; all course questions are original.

AASA sample item references: 18

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01 / Learn

The base is the repeated factor and the exponent tells how many copies to multiply. For a positive exponent, 4^3 = 4 × 4 × 4, not 4 × 3. Parentheses show when a whole product is the base.

Worked example 1

Write 3^2 as repeated multiplication and evaluate.

1. The base is the repeated factor and the exponent tells how many copies to multiply.

2. For a positive exponent, 4^3 = 4 × 4 × 4, not 4 × 3.

3. The exponent counts 2 factors of 3

4. multiply all factors.

5. State the result in the form requested: 3 × 3 = 9.

Answer: 3 × 3 = 9

Worked example 2

Select all expressions equal to 2^3.
A. 9
B. 2 × 3
C. 8
D. 2 × 2 × 2

1. The base is the repeated factor and the exponent tells how many copies to multiply.

2. The exponent 3 means three factors of 2

3. their product is 8.

4. It does not mean 2 times 3.

Answer: C, D - 8; 2 × 2 × 2

Day 046 | Your turn

Work in order: foundation → application → explanation. Show calculations and label units.

01 / Constructed response

Write 3^3 as repeated multiplication and evaluate.

02 / Constructed response

Write 2^4 as repeated multiplication and evaluate.

03 / Constructed response

Write 4^3 as repeated multiplication and evaluate.

04 / Constructed response

Write 6^4 as repeated multiplication and evaluate.

05 / Constructed response

Write 7^2 as repeated multiplication and evaluate.

06 / Constructed response

Write 2^2 as repeated multiplication and evaluate.

07 / Constructed response

Write 4^3 as repeated multiplication and evaluate. Explain why it differs from 4 × 3.

08 / Multiple select

Select all expressions equal to 3^3.
A. 3 × 3
B. 27
C. 28
D. 3 × 3 × 3

09 / Multiple select

Select all expressions equal to 6^3.
A. 6 × 3
B. 6 × 6 × 6
C. 217
D. 216

10 / Constructed response

Write 3^3 as repeated multiplication and evaluate. Explain why it differs from 3 × 3.

Day 046 | Unit 04 | Answer key

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1. 3 × 3 × 3 = 27

The exponent counts 3 factors of 3; multiply all factors.

6.EE.A.1 | Constructed response

2. 2 × 2 × 2 × 2 = 16

The exponent counts 4 factors of 2; multiply all factors.

6.EE.A.1 | Constructed response

3. 4 × 4 × 4 = 64

The exponent counts 3 factors of 4; multiply all factors.

6.EE.A.1 | Constructed response

4. 6 × 6 × 6 × 6 = 1296

The exponent counts 4 factors of 6; multiply all factors.

6.EE.A.1 | Constructed response

5. 7 × 7 = 49

The exponent counts 2 factors of 7; multiply all factors.

6.EE.A.1 | Constructed response

6. 2 × 2 = 4

The exponent counts 2 factors of 2; multiply all factors.

6.EE.A.1 | Constructed response

7. 4 × 4 × 4 = 64; 4 × 3 = 12

The exponent counts 3 factors of 4; multiply all factors.

6.EE.A.1 | Constructed response

8. B, D - 27; 3 × 3 × 3

The exponent 3 means three factors of 3; their product is 27. It does not mean 3 times 3.

6.EE.A.1 | Multiple select

9. B, D - 6 × 6 × 6; 216

The exponent 3 means three factors of 6; their product is 216. It does not mean 6 times 3.

6.EE.A.1 | Multiple select

10. 3 × 3 × 3 = 27; 3 × 3 = 9

The exponent counts 3 factors of 3; multiply all factors.

6.EE.A.1 | Constructed response